

Air Traffic and Usage Predictions in Avionic Communications using Attention Based VAEGAN Model

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Abstract—The need for uninterrupted connection is an enabler for enhanced connectivity in aircrafts. Satellite based communication in aircrafts exhibits high latency and can have limited data rates. Furthermore, the increasing demand for air travel can strain the capacity of satellite communication systems, necessitating the development of more robust traffic prediction methods. This necessity is particularly pronounced in the realm of business aviation, given the irregular traffic patterns compared to scheduled commercial flights. In this paper, we present an attention-based VAEGAN model designed to forecast the number of active tails within satellite beams. We extend the capabilities of our proposed model to predict the volume of upstream and downstream usage within these satellite beams. To validate our model, we employ real avionics data collected from the two most heavily traversed flight routes. Finally, we perform a comparative analysis, benchmarking the performance of existing machine learning-based techniques with our proposed model. The findings indicate that the proposed VAEGAN model exhibits superior performance in forecasting irregularities in the time-series pattern, specifically in forecasting unusual highs or lows in the number of aircrafts within the satellite beam, outperforming alternative models.

Index Terms—Avionics, Machine Learning.

I. INTRODUCTION

In 2019, there was a total of 40 million flights around the world, with the busiest day having 230,000 flights [1]. A typical flight will generally generate several gigabytes of data while next generation flights can generate 7 to 8 terabytes worth of data. With the growing number of annual flights, and enhanced connectivity, it is estimated that annual flight data generation will reach 98 million terabytes by 2026 [1]. Leveraging such data can not only prove to be profitable, but can also enrich the overall flight experience from flight scheduling, delays, boarding, in-cabin experience and aircraft health. As the avionics industry is rapidly growing, so is the need for increased in-cabin bandwidth, differentiated connectivity services, enhanced air traffic control and faster maintenance [2].

The requirement for permanent connection for aircrafts, airports, air traffic control and maintenance systems along with the introduction of aircraft-to-aircraft communication links is a primary enabler for enhanced connectivity in such systems [2]. Moreover, in-cabin connectivity tends to be much slower than the usual connectivity achieved in terrestrial networks. This is due to the limited capacity achieved through aircraft communication systems in comparison to terrestrial networks, which leaves room for improvement and optimization in this domain. In-cabin connectivity can be achieved through two primary methods. The first method involves direct air-to-ground communication, where the aircraft's terminal communicates directly with ground-based stations. The second method utilizes satellite-based communication, where the aircraft terminal communicates with satellite constellations that, in turn, relay data to and from ground stations to access the internet. Both communication schemes pose their own set of challenges which can lead to some limitations. For instance, satellite based communication exhibits higher latency, has limited data rates and can be costly [3]. On the other hand, direct air to ground communication has limited coverage and is only feasible during take-off, landing, and continental flights. In this paper, we focus on improving satellite based connectivity on aircrafts as we aim to predict traffic and usage patterns within various satellite coverage regions in order to optimize aircraft connectivity.

Satellite communication systems typically consist of multiple spot beams, formed by a single satellite, over a region of interest. Each beam targets a point on the earth's surface and has its own set of performance indicators, such as power, capacity, data rate, etc [4]. The performance of the beam is indicated by the satellite provider, and is typically selected based on the demand in the region. Having prior knowledge of the number of flights utilizing bandwidth within each beam, as well as the volume of bandwidth consumption can assist satellite providers with appropriately allocating resources to the beams. Moreover, this can also allow the

flight to optimize the in-cabin connectivity by either deviating the flight path slightly, which can typically be done in non-commercial flights, or switching between satellite providers to avoid congested beams. The concept of dynamic network switching for throughput optimization has been studied in [5] and [6]. Leveraging real time traffic predictions is also an essential foundation for bandwidth management in avionics communication, a concept explored in [7].

The challenge described above is complex and often non-linear, which can be difficult to solve through mathematical modelling. Moreover, this challenge becomes particularly interesting in business aviation as traffic patterns tend to be more erratic, unlike commercial aviation where flights are typically scheduled months in advance [8]. Application of machine learning techniques allows for the use of the millions of terabytes of flight data that has been collected throughout the years, while providing a solution for the challenge mentioned above. Machine learning has been widely implemented in applications such as speech, image recognition and financial predictions. Within recent years, artificial intelligence has also made a breakthrough within the telecommunications industry and has been applied to problems such as routing, resource allocation and localization. While the application of machine learning in the avionics industry is still in its early stages, recent research endeavors have demonstrated its potential to deliver cutting-edge outcomes. Notably, machine learning techniques have been investigated in various avionic domains, including path loss prediction and signal modulation classification within aircraft cabins [9], [10]. Additionally, machine learning has been explored for packet routing optimization in satellite networks [11]. These endeavors collectively highlight the promising prospects of machine learning in enhancing avionics technologies.

In this paper we propose an attention-based Variational Auto-Encoder General Adversarial Network (VAEGAN) model to predict the traffic within beams - both in terms of the number of active flights and usage. Our contributions are summarized below:

- We propose an attention-based VAEGAN model to predict the number of active tails in each beam. We define active tails as aircraft's that are connected to the satellite network
- We extend our model to predict the total usage in each beam
- We test the performance of our model using a real world dataset in the two busiest flight routes - across the continental United States (CONUS) and trans-Atlantic flights
- We benchmark the performance of our model against other state-of-the-art time series prediction models

In Section II, we introduce prior research that leverages machine learning-based time series forecasting techniques. These studies serve as benchmark models for evaluating the performance of our proposed model, which is described in greater detail in Section V. Section III offers a comprehensive

description of the dataset utilized in our study. In Section IV, we provide an in-depth explanation of our proposed model. Lastly, in Section VI, we outline the future directions and next steps for our research study.

II. RELATED WORK

Machine learning based avionics communication is a new and emerging field with limited studies. However, time series forecasting and predictions is a field of study that predates the introduction of machine learning by many years. In this paper, we draw inspiration from prediction algorithms employed in addressing air traffic congestion. While machine learning-based avionics communication represents a new area with limited prior research, it's essential to recognize the significance of time series forecasting. This foundation in forecasting techniques not only underpins our exploration of prediction algorithms but also allows us to draw inspiration from addressing the challenges posed by air traffic congestion, a phenomenon closely intertwined with potential satellite beam congestion. The first section of our literature review focuses on several notable studies that have investigated the application of advanced techniques such as Long Short-Term Memory (LSTM), k-Nearest Neighbors (KNN), and Support Vector Regression (SVR) in the context of air traffic flow value predictions. Additionally, we introduce some studies that employ time series forecasting methods in the non-avionics space, encompassing areas such as stock market predictions, sensor predictions, and more, highlighting the versatility and applicability of these approaches.

The use of LSTM networks, SVR, K-nn and various clustering and regression models has been studied and proved to achieve promising results in terms of predicting traffic flow values and arrival and departure delays. The work in [12] proposes two prediction models to predict air traffic flow. The first model is an LSTM based model and the second model is a Support Vector Regression (SVR) based model. The model takes the hour of the day, day of the week, day of the month, season of the year, type of day (statutory holiday, weekend or weekday), positional data and the historical average traffic flow as an input and outputs a predicted air traffic flow value. Their results show that the LSTM predictor outperforms the SVR predictor as the predictions received an overall smaller average error with respect to the actual traffic flow value.

The authors in [13] propose a K-Nearest Neighbour (K-nn) regression model for air traffic flow prediction. K-nn is a simple regression model in which the label of the test sample is predicted based on the test sample's distance to other training samples. The authors use real air traffic flow data from 4 corridors in a Beijing terminal and compare the results of the of their proposed technique to the SVR model. The results show that both the K-nn and SVR models are capable of accurately predicting the flow value changes. While both models cannot predict irregular fluctuations of the actual flow values, the prediction curves of K-nn are smoother and coincide with the actual flow values.

[14] proposes a method to predict uncertainty in the four-dimensional positions of aircrafts using machine learning in order to improve trajectory estimation. The paper derives a performance predictability approach using clustering and regression. Different clustering techniques such as K-means, density-based clustering, farthest first and Cobweb are considered. 5 different predictors are designed: 2-dimensional deviation model, delay threshold forecast model, delta flight level forecast model, predictability metric forecast model and cluster ranking forecast model. Their results show that delay forecasting and cluster ranking predictions achieve promising results.

[15] proposes a Variational Autoencoder Conditional Generative Adversarial Network (VAEGAN) model for long term multi-variate time series predictions. They test their proposed model using five real world datasets that include stock market, energy and electroencephalogram (EEG) data. The VAE component helps in learning the latent representations of the time series data, while the GAN component refines these representations to generate long-term forecasts. Their results show that the use of generative networks such as VAEGANS can outperform more commonly used models such as LSTMs.

The studies summarized above demonstrate the potential machine learning techniques when it comes to predicting air traffic flow. LSTM networks particularly, have been able to achieve minimal errors, as shown in the work done in [12]. This is likely due to the chronological nature of flight data, as the number of flights greatly varies depending on the day, time and season. LSTM networks are capable of modeling such long-term time dependencies, thus, providing more accurate results. However, more improvement in this domain can be made by investigating abnormal changes in traffic flow. Flight trajectories can sometimes be unpredictable, and relying only on historical data can limit the model's capabilities. [15] showcases how generative networks like VAEGAN can be applied not just for data generation but also for achieving precise time series predictions. While they showcase the adaptability of their approach through experimentation with diverse datasets, it's important to note that none of the datasets they tested accurately reflect the conditions within the avionics communication domain. In this paper, we build upon the findings presented in [12], establishing a connection between air traffic flow value and satellite beam congestion. This connection helps us identify a valuable set of input features for model training. Additionally, we draw inspiration from [15], where we propose a modified attention-based VAEGAN model to suit our dataset, enabling us to make predictions regarding satellite beam congestion.

III. DATASET

We utilize a real dataset that comprises positional and network performance information for business aviation jets. Specifically, the dataset contains longitude, latitude, timestamps, satellite name, beam name, a flag indicating whether the aircraft's terminal is connected to the satellite network, signal to noise ratio (SNR), downstream allocation to demand,

and usage (MB). The dataset contains data from January of 2020 to August of 2023. We consider a Geostationary (GEO) satellite constellation, operating in the Ka-band. This choice of frequency enables higher data rates but results in narrower satellite beams as a trade-off [3]. For the purpose of the study, we only select beams and satellites from the two busiest flight routes - across the continental United States and Trans-Atlantic flights. For the remainder of the paper, we will refer to the satellites as Satellite A and Satellite B. The dataset contains 3 beams from Satellite A and 12 beams from Satellite B, which we will refer to as beams 1-A/B, 2-A/B, and so on. Figure 1 displays an example of the placement of the satellites and beams, with labels.

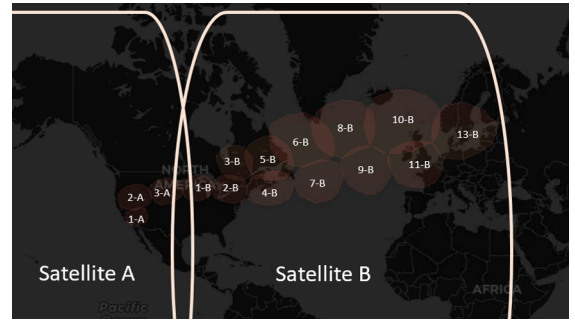


Fig. 1. Image showcasing satellite and beams placement

To preprocess the data, we initially organized it chronologically by date. Subsequently, we augmented the dataset with season indices and holiday indicators to determine the prevailing season during each flight and whether the day was a statutory holiday. We consider all US federal holidays. Further data parsing involved extracting the year, month, day of the month and day of the week from the date. The data was then segmented into 15-minute intervals, during which we calculated the count of unique flights associated with each beam. We additionally compute the mean historical Signal-to-Noise Ratio (SNR) and downstream allocation to demand for each beam within the specified time window. These metrics play a pivotal role in identifying congestion-related issues. A higher SNR allows for more reliable communication with lower error rates. When SNR is high, it is possible to allocate resources more efficiently and transmit data at higher data rates, meeting the demands of users for high-quality connections and faster data transfer. On the other hand, congestion occurs when the demand for network resources exceeds the available capacity. A high SNR can enable efficient resource allocation, which, in turn, can help mitigate congestion [16]. Hence, we can deduce that a combination of high SNR and low allocation to demand can suggest a congestion problem. Conversely, a scenario characterized by a low SNR paired with low allocation to demand could suggest a Radio Frequency (RF) performance issue. Additionally, it's worth noting that factors like interference and power levels can also influence network performance. However, our research has demonstrated that incorporating SNR and allocation to demand as input

features in our model has proven effective in enhancing overall performance in terms of predicting air traffic within the beam. In the extended version of the model, which forecasts usage per beam, an additional step involved aggregating the total usage for each beam within that specific time window. The data is then normalized using min-max normalization.

The last phase involves dividing the dataset into input and output components, as well as partitioning it into training and testing sets. Given the time-series forecasting nature of our proposed approach, we organize the inputs and outputs such that the information from the preceding four timesteps is used to predict the subsequent timestep. Specifically, considering our data is segmented into 15-minute intervals, we utilize data from the past hour to predict the next 15 minutes. This choice is based on our assessment of an appropriate time window, considering factors such as the coverage area of the satellite beams and the velocity of the aircrafts. The inputs of the models are summarized below:

- **LSTM Model Predicting Number of Active Tails:** Year, Month, Day of Month, Day of week, Hour, Minute, Average Allocation to Demand, Average SNR, Holiday Index, Season Index, Average Air Traffic Flow Value, Flight Count Per Beam
- **LSTM Model Predicting Usage:** Year, Month, Day of Month, Day of week, Hour, Minute, Average Allocation to Demand, Average SNR, Holiday Index, Season Index, Average Air Traffic Flow Value, Flight Count Per Beam, Total Usage per Beam in MegaBytes (MB)
- **VAEGAN Model Predicting Number of Active Tails:** Flight Count Per Beam
- **VAEGAN Model Predicting Usage:** Total Usage Per Beam in MB

IV. PROPOSED TECHNIQUE

In this paper, we propose an attention-based VAEGAN model to predict the number of active tails within a satellite beam. We also extend our model to predict the total usage per beam, by utilizing the same architecture and varying the training data as described in section III. The model consists of an encoder, decoder and discriminator, as shown in Figure 2.

Furthermore, we deploy a basic LSTM model and an attention-based LSTM model as benchmark models to assess and compare the outcomes against the results of our proposed attention-based VAEGAN model. The LSTM model consists of a single LSTM layer with 16 units, and a fully connected output layer with 16 units. Meanwhile, the attention-based LSTM model consists of a single LSTM layer with 16 units followed by an attention layer and a fully connected output layer with 16 units. Both models are trained using the data described in section III.

A. Model Architecture

1) **Encoder:** The encoder comprises two key components: a multi-head attention layer and an LSTM layer. The input to the encoder is the input feature vector described in Section III. The

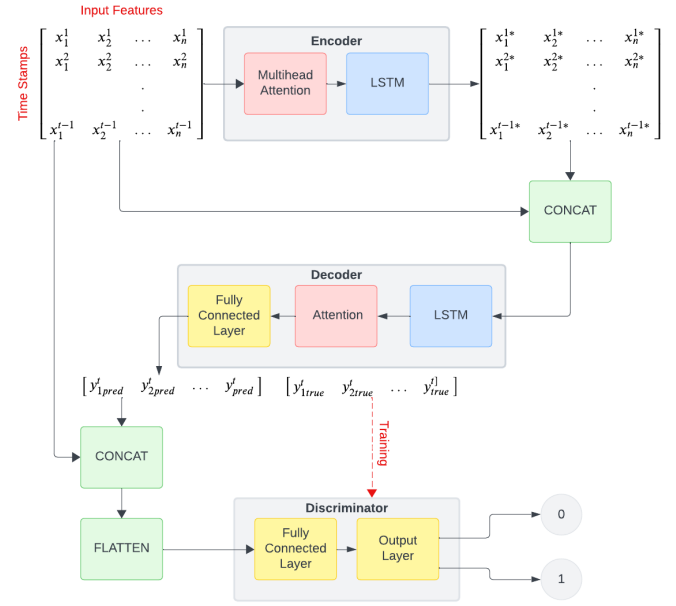


Fig. 2. Block Diagram of Proposed Model

attention mechanism enables the model to emphasize specific segments of the input data during output generation, facilitating the capture of long-range dependencies and relationships within sequences. Alignment scores are computed for each element in the input sequence, which represent the importance of each element with respect to the desired output. These alignment scores are subsequently transformed into attention weights through a softmax function. The attention weights are then used to calculate a context vector, which is essentially a weighted sum of the input elements [17]. In our model, we use an extension of the attention mechanism, called multi-head attention, which enables the model to concurrently focus on different data portions. We also employ self-attention, where the attention weights are trained based on the input itself, allowing the model to capture inherent relationships within the data. The context vector serves as input to the LSTM network. The LSTM network returns a matrix of the same dimensions as the input matrix, represented as X^* in Figure 2.

- **Multihead Attention:** Number of heads = 4
- **LSTM:** units = Number of features, Return Sequences = True

2) **Decoder:** The decoder consists of an LSTM layer, an attention layer and fully connected layer. The input to the decoder is a matrix consisting of the output of the encoder, X^* , and the input feature matrix, X , described in section III. In the decoder, we use cross-attention, where the computation of attention scores is based on the output. The context vector is then used as an input to the fully connected layer, which outputs the predictions, Y_{pred} .

- **LSTM:** units = Number of features, Return Sequences = True, activation = ReLU

- **Dense Layer:** units = 16, activation = Sigmoid

3) *Discriminator:* The discriminator consists of a fully connected layer and an output layer. The primary purpose of the discriminator is to assess the authenticity of the data produced by the generator. Hence, the input to the discriminator is a matrix containing the input sequence, X , and the output at timestamp t , Y_{pred} or Y_{true} . The output of the discriminator is a probability score indicating how likely the data it was fed, is real or fake. Subsequently, it outputs a 1 if the data is likely to be real and a 0 otherwise.

- **Dense Layer:** units = 100s , activation = ReLU
- **Dense Layer:** units = 1, activation = Sigmoid

B. Model Training and Implementation

We implemented our proposed model using Python and TensorFlow Functional API, leveraging TensorFlow's built-in *Attention* and *MultiHeadAttention* classes for the attention mechanisms. Our training process is outlined in Algorithm 1.

Algorithm 1 Prediction Algorithm

Data: Active Tail Count, Total Usage [t-1, t-2, t-3, t-4]

Result: Active Tail Count, Total Usage [t]

Initialize encoder, decoder and discriminator

for n epochs **do**

 encode input into latent space via encoder

 Predict output, y_{pred}^t , via decoder

 Determine probability that y_{pred}^t is real via discriminator

 Determine probability that y_{true}^t is real via discriminator

 Calculate generator and discriminator loss

end

We partitioned our dataset into training, validation, and testing sets, employing a learning rate of 0.0001. Throughout this section, we collectively refer to the encoder and decoder as the "generator."

To make predictions, we utilized the generator with training inputs. These predictions, denoted as y_{pred}^t were concatenated with the previous timesteps, $[x^{t-1}, x^{t-2}, x^{t-3}, x^{t-4}]$, forming a complete sequence. This sequence was then fed into the discriminator. The discriminator's output is a probability score of either 1 or 0, indicating whether the sequence is real or fake. In parallel, we constructed another sequence using the actual output value, y_{true}^t , and supplied it to the discriminator for evaluation. These steps are repeated for each iteration of the training process. At the end of each iteration, we compute the generator and discriminator losses to assess the performance and convergence of the model. The generator loss is shown in Equation 4, and is a weighted sum of the Mean Squared Error (MSE), Equation 1, Mean Absolute Error (MAE), Equation 2, and binary cross entropy loss, Equation 3. The values for the weights $l1$, $l2$, and $l3$ were selected through an iterative process. The discriminator loss is shown in Equation 5 and is a combination of the binary cross entropy of the true and predicted outputs.

$$MSE = \frac{1}{n} \sum_{n=1}^{totalsamples} (y_{true} - y_{pred})^2 \quad (1)$$

$$MAE = \frac{\sum_{n=1}^{totalsamples} |y_{true} - y_{pred}|}{n} \quad (2)$$

$$loss(a, b) = \max(b, 0) - ba + \log(1 + e^{-|b|}) \quad (3)$$

$$generatorLoss = l1 * MSE + l2 * MAE + l3 * loss(1, y_{pred}) \quad (4)$$

$$discriminatorLoss = loss(1, y_{pred}) + loss(1, y_{true}) \quad (5)$$

V. PERFORMANCE EVALUATION

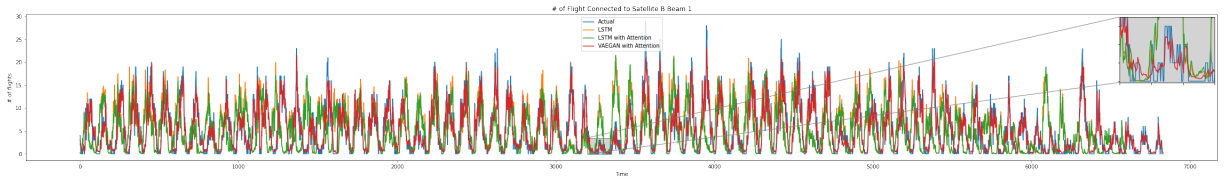
In order to evaluate the effectiveness of our proposed model, we use the dataset outlined in Section III. Furthermore, we establish two benchmark models for comparison: a traditional LSTM model and an attention-based LSTM model. The initial segment of this section concentrates on assessing the performance of our proposed model for predicting the number of active tails within a beam. Subsequently, we consider evaluating the model's proficiency in predicting the total downstream usage within the beam.

We utilize two metrics, RMSE (Root Mean Square Error) and the R-squared score (R^2), to evaluate the performance of our models. RMSE quantifies the average error magnitude between predicted and actual values. Since we are predicting the number of active tails, RMSE is unitless but signifies the prediction error in terms of the number of flights. Conversely, the R^2 score measures the proportion of variance explained between actual and predicted values. An R^2 score closer to 1 indicates a better alignment between the true and predicted values [18].

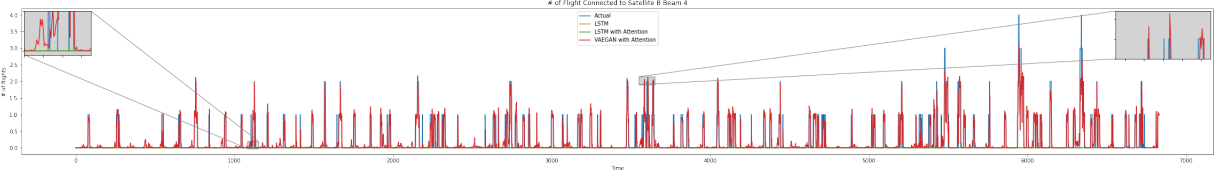
Before delving into a detailed analysis of the results, we will classify the beams from our dataset into three distinct categories: high-traffic beams (maximum flights >10), mid-traffic (5-10 flights) and low-traffic beams (< 5 flights). Here, we define the number of flights as the maximum number of flights observed per beam, in the dataset. We choose to classify the beam traffic based on the maximum, rather than the average, as the performance of the different models varies based on their ability to predict irregular peaks and valleys in the sequence. This categorization will provide valuable insights into the models' performance across different traffic scenarios. The beam classification is outlined as follows:

- **High Traffic Beams:** 1-A, 3-A, 1-B, 2-B
- **Mid Traffic Beams:** 2-A, 3-B, 5-B, 6-B, 7-B, 8-B, 9-B
- **Low Traffic Beams:** 4-B, 10-B, 11-B, 12-B

Table I presents a summary of the results for predicting the number of active tails within a beam. Notably, the performance of the LSTM model benefits from the inclusion of the attention mechanism. Additionally, it is observed that all models exhibit favorable performance in high-traffic beams. In most cases, the

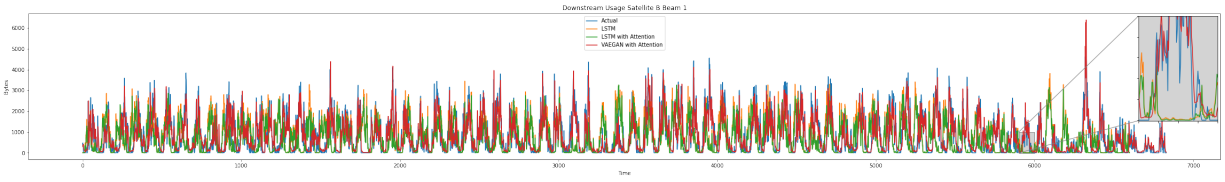


(a)

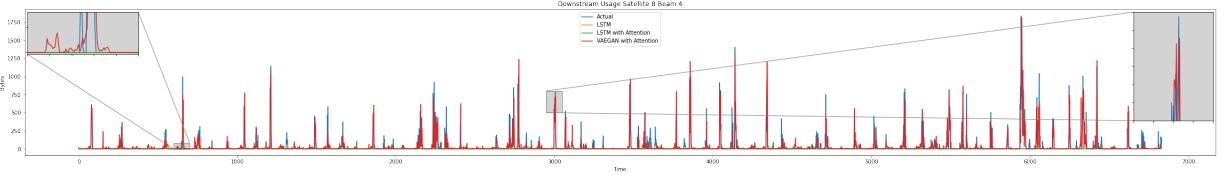


(b)

Fig. 3. (a) Air Traffic predictions for Beam 1-B, demonstrating high traffic beam (b) Air Traffic Predictions for beam 4-B, demonstrating low traffic beam



(a)



(b)

Fig. 4. (a) Usage for Beam 1-B, demonstrating high traffic beam (b) Usage for beam 4-B, demonstrating low traffic beam

RMSE is less than 1, which can be considered insignificant especially in high traffic beams. Figure V-a presents a sample plot illustrating the prediction outputs of all models. This visual representation vividly demonstrates that the attention-based VAEGAN model excels in predicting irregularities within the time-series pattern when compared to the LSTM and attention-based LSTM models. We outlined an instance where there was an unexpected drop in the total number of flights. The attention-based VAEGAN model was accurately able to detect this drop, whereas other models could not.

Furthermore, a significant improvement in prediction accuracy becomes apparent in low-traffic beams when employing the attention-based VAEGAN model. This becomes evident when examining cases where the RMSE is low, but the R^2 score still indicates suboptimal alignment between actual and predicted series. A closer examination is provided in Figure V-b, where it is evident that both the LSTM and attention-based LSTM models struggle to make accurate predictions for low-traffic beams. Their predictions result in a flat line, effectively forecasting zero flights for every instance. Conversely, the attention-based VAEGAN model consistently maintains strong

performance for both high-traffic and low-traffic beams, showcasing its robustness across varying traffic conditions. This can be examined further in Table I, beams 4-B, 10-B and 11-B. By observing mid traffic beams, such as 6-B and 7-B, we can also see that adding the attention mechanism to the LSTM model improves performance and allows the model to continue to make accurate predictions as the number of active tails decreases.

Table II summarizes the results for the extended version of the model, where we predict downstream usage for each beam. Similar performance is also observed, where all models perform well when it comes to predicting usage in high and mid traffic beams. The attention-based VAEGAN model outperforms other models when it comes to predicting usage in low traffic beams.

VI. CONCLUSION

Having prior knowledge of the number of flights utilizing bandwidth within each satellite beam, as well as the volume of bandwidth consumption, allows flights to optimize in-cabin connectivity. This optimization may involve strategic

TABLE I

TABLE SUMMARIZING MODEL PERFORMANCE FOR FLIGHT TRAFFIC PREDICTIONS

Beam	RMSE			R2		
	LSTM	LSTM w/ Att.	VAEGAN w/ Att.	LSTM	LSTM w/ Att.	VAEGAN w/ Att.
1-A	1.05	1.08	1.15	0.85	0.85	0.83
2-A	0.71	0.72	0.71	0.70	0.69	0.70
3-A	0.88	0.87	0.92	0.82	0.83	0.82
1-B	1.57	1.68	1.70	0.90	0.89	0.89
2-B	1.07	1.06	1.20	0.85	0.85	0.81
3-B	0.59	0.59	0.58	0.57	0.56	0.57
4-B	0.49	0.49	0.25	-0.16	-0.16	0.70
5-B	0.76	0.75	0.78	0.65	0.65	0.81
6-B	1.07	0.53	0.59	-0.58	0.64	0.58
7-B	0.93	0.45	0.55	-0.44	0.64	0.50
8-B	1.35	0.55	0.56	-0.7	0.72	0.71
9-B	1.71	0.66	0.67	0.85	0.72	0.71
10-B	0.42	0.42	0.26	-0.14	-0.14	0.54
11-B	0.29	0.29	0.17	-0.06	-0.06	0.81
12-B	0.43	0.43	0.23	-0.106	-0.106	0.67
All	2.51	2.49	2.66	0.96	0.96	0.95

TABLE II

TABLE SUMMARIZING MODEL PERFORMANCE FOR DOWNSTREAM USAGE PREDICTIONS

Beam	RMSE			R2		
	LSTM	LSTM w/ Att.	VAEGAN w/ Att.	LSTM	LSTM w/ Att.	VAEGAN w/ Att.
1-A	319	297	300	0.70	0.74	0.74
2-A	179	180	184	0.53	0.53	0.50
3-A	242	243	260	0.71	0.71	0.67
1-B	403	420	457	0.79	0.77	0.73
2-B	272	276	286	0.73	0.72	0.70
3-B	130	132	126	0.37	0.41	0.44
4-B	129	129	96	-0.07	-0.07	0.4
5-B	129	184	214	0.49	0.47	0.30
6-B	220	152	134	-0.27	0.38	0.52
7-B	177	125	114	-0.18	0.40	0.50
8-B	257	158	147	-0.33	0.49	0.56
9-B	280	172	167	-0.35	0.48	0.52
10-B	86	86	65	-0.04	-0.04	0.38
11-B	58	58	44	-0.02	-0.02	0.37
12-B	699	738	742	0.91	0.90	0.90

decisions such as adjusting the flight path, Intelligent routing on the LAN, or switching between satellite providers to circumvent congested beams and ensure uninterrupted and efficient connectivity for passengers and operations. In this paper, we introduce an attention-based VAEGAN model for predicting the number of active tails within each satellite beam. Additionally, we extend our proposed technique to forecast upstream and downstream usage within each beam. To gauge the effectiveness of our approach, we employ LSTM and attention-based LSTM models as benchmarks. Our findings reveal that while all models perform well for high-traffic beams, the attention-based VAEGAN model exhibits substantial improvements for low-traffic beams, showcasing its versatility and proficiency in predicting irregularities. This adaptability is pivotal as it facilitates scalability, allowing a single model to predict air traffic and usage volumes for all beams within a satellite network.

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